

InsightPilot: An Agentic Platform for Conversational Business Intelligence

Project Abstract

by

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ABSTRACT

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Enterprise organizations depend on data analysts to answer unanticipated questions, from requests for new metrics to ad hoc comparisons. These analysts locate the data sources, inspect the schema, write queries, and select suitable visualizations. Business Intelligence automation, as implemented in Microsoft Power BI Copilot, Databricks Genie, and Snowflake Cortex Analyst, allows such questions to be converted automatically into SQL.

These tools are powerful, but they are also confined to a single pre-defined warehouse and unable to span multiple systems. Worse, they are confidently wrong at times. As Omni Analytics found in its 2026 analysis of 4,602 faulty text-to-SQL queries, 81.2% of these errors were caused by a wrong column or a misread meaning. When the output looks sufficiently plausible, analysts are forced to rebuild queries to be sure. As a 2026 dbt Labs survey of 363 practitioners found, 71% ranked incorrect output among their leading concerns. Analysts need answers that can span their real sources and cost less to check than to rebuild.

In this project, we build InsightPilot, an agentic platform that delivers conversational business intelligence across real data sources. An orchestrator turns a question into an analysis plan that the analyst reviews before any query runs. Four specialist agents then fetch data from relevant sources, analyze it, model it when the question is predictive, and present the result. One investigation can draw on relational databases, spreadsheets, and web services. Every step records the source, query, and rows returned. A separate check confirms that the step matched the plan, yielding an answer the analyst can verify by referring to the plan without rebuilding it.